



Comparative Analysis of Tea Plucking Harvesters: Evaluation of Existing Systems and Development of Improvement Recommendations

ME325- Mechanical Engineering Group Project

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Content....

1. Introduction
 - a. Background
 - b. Purpose and Need of the Project
2. Analysis of Existing Systems
 - a. Technical Specifications of Existing Tea Harvesting Machines
 - b. Performance Comparison of Tea Harvesting Machines
 - c. Comparative Rating of Tea Harvesting Machines
3. Project Objectives & Methodology
 - a. Project Objectives
 - b. Project Methodology
4. Project Planning & Timeline
5. Impact Analysis
 - a. Economic Considerations
 - b. Social Considerations
 - c. Environmental Considerations
6. References

1. Introduction

1.1 Background

Tea is a major agricultural crop in Sri Lanka and makes an important contribution to the national economy through export income, employment, and rural development. Sri Lanka is well known worldwide for producing high-quality Ceylon tea. Tea harvesting is one of the most labour-intensive tasks in tea production, as plucking must be carried out regularly while maintaining strict quality standards. Traditionally, tea leaves are harvested manually by skilled workers who carefully select tender shoots to ensure better quality and future plant growth.

However, the tea industry is currently facing major challenges such as labour shortages, increasing wages, and high production costs, making manual plucking less sustainable. To address these issues, mechanical tea harvesting systems such as handheld shears, motorized harvesters, and portable tea plucking machines have been introduced to improve efficiency and reduce dependence on labour. Although these technologies can increase harvesting speed, they also create problems including leaf damage, poor shoot selection, inclusion of coarse material, and operational difficulties in uneven field conditions.

A major challenge in mechanical tea harvesting is maintaining the required plucking standard. In tea cultivation, the harvestable portion is called the pluckable shoot, which generally consists of a bud with two or three tender leaves. The plucking standard varies depending on climatic conditions and plantation management practices. During wet seasons, rapid plant growth allows comparatively harder plucking standards such as three leaves and a bud, while during dry seasons lighter plucking standards such as two leaves and a bud are preferred to maintain tea quality.

Understanding tea shoot structure is essential when analyzing mechanical tea harvesting systems because harvesting efficiency and quality depend on correctly identifying pluckable shoots. A tea shoot consists of the apical bud, which is the actively growing terminal bud responsible for shoot elongation, followed by progressively developing young leaves. Near the shoot base, the first immature protective leaf is called the fish leaf, while the mature supporting leaf left after harvesting is known as the mother leaf. Very young immature shoots not ready for harvesting are locally called Arimbu and must remain until reaching the required plucking stage. Tea buds can be classified as active buds and dormant buds, depending on growth conditions, while some shoots may enter a temporary resting condition called banji, where the apical bud becomes inactive and shoot growth stops, making them unsuitable as ideal pluckable shoots. After harvesting, new shoots regenerate from remaining buds, with growth influenced by temperature, rainfall, sunlight, and soil moisture. Tea bushes are maintained at a uniform height known as the plucking table, and the time interval between two harvesting cycles is called the plucking round, both of which are important for maintaining sustainable yield and tea quality.

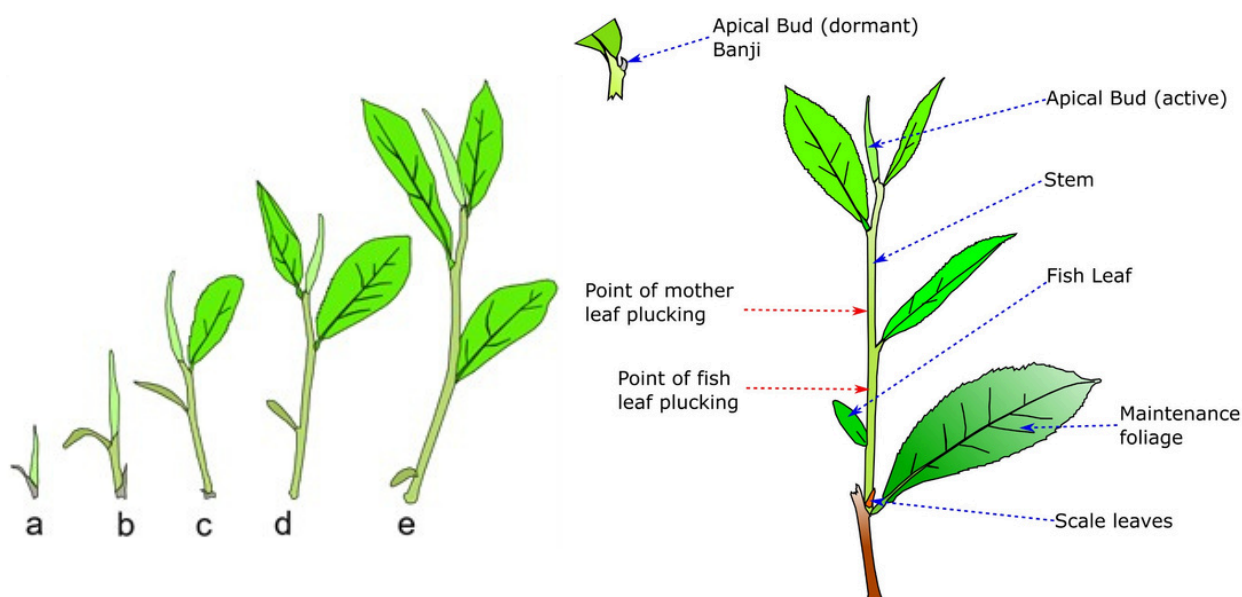


Figure 01: Tea shoot structure and key components relevant to plucking.

Tea-growing regions can generally be divided into tropical regions and temperate regions according to climatic conditions. Sri Lanka belongs to the tropical region, where relatively warm temperatures and seasonal rainfall allow tea plants to grow continuously throughout most of the year. Because of this continuous growth, tea bushes in Sri Lanka require frequent plucking rounds, often within 7–14 days depending on weather conditions and elevation. In contrast, tea-growing countries in temperate regions experience colder seasons and slower plant growth, resulting in longer plucking intervals and seasonal harvesting patterns.

Climatic and field condition differences create major challenges when applying tea harvesting machines developed for temperate regions to Sri Lankan plantations. These machines are generally designed for uniform terrain, slower shoot growth, and longer plucking rounds, whereas Sri Lankan tea fields have steep slopes, uneven terrain, dense shoot growth, and shorter plucking cycles. As a result, many imported machines show poor adaptability and cannot maintain accurate selective plucking standards. A major limitation is the inability to distinguish between pluckable shoots and non-pluckable shoots such as Arimbu, banji shoots, and other immature developing shoots, causing machines to harvest both required and non-required material simultaneously. This reduces tea quality, damages future yield potential, increases the inclusion of coarse leaves and stems, and, due to uneven terrain, makes it difficult to maintain a constant cutting height, resulting in inconsistent harvesting performance and increased operator fatigue.

1.2 Purpose and Need of the Project

Mechanical tea harvesting systems have been introduced as a possible solution to reduce labour dependency and increase harvesting productivity. Although these systems improve harvesting speed, many existing machines are unable to satisfy the quality requirements.. The major technical issue is the lack of selective plucking capability. Existing machines cannot properly differentiate between mature pluckable shoots and immature shoots such as Arimbu, active buds, and developing shoots intended for future plucking rounds. Consequently, machines remove both required and non-required shoots together, reducing final tea quality and affecting future yield.



Figure 02: Tea Bush Surface after Mechanical Harvesting Showing Non-Selective Cutting Pattern



Figure 03: Field Observation at TRI Showing Tea Bush Damage and Immature Shoot Loss Caused by Existing Mechanical Harvesting Systems

Another major problem is the difficulty of operating mechanical harvesters on uneven terrain. Tea plantations in Sri Lanka are commonly located on steep hillsides and irregular land surfaces. Many available machines are designed for relatively flat tea fields found in temperate regions and therefore cannot maintain a stable cutting height under local plantation conditions. This results in inconsistent plucking, increased leaf damage, operator discomfort, and lower harvesting accuracy.

In addition, differences between tropical and temperate tea-growing conditions significantly affect machine performance. Due to faster shoot growth in tropical climates, Sri Lankan tea bushes require shorter plucking rounds and more accurate harvesting control. Machines developed for temperate regions may not satisfy these requirements because they are designed for different shoot densities, growth patterns, and harvesting intervals.

Therefore, this project is necessary to conduct a detailed evaluation of currently available tea harvesting systems and identify their limitations under Sri Lankan plantation conditions. The findings of this study can contribute toward the development of more suitable harvesting solutions capable of improving harvesting efficiency while maintaining tea quality standards. The project will also provide engineering recommendations for future design improvements focusing on selective plucking capability, terrain adaptability, ergonomic operation, and reduced leaf damage.

2. Analysis of Existing Systems

Our team conducted a field visit to the Tea Research Institute (TRI) sub-center in Ratnapura to gather primary data and observe operational benchmarks. During this visit, we conducted a comparative observation of traditional manual plucking and current mechanical systems, specifically focusing on how manual labor achieves high selectivity by identifying mature "two leaves and a bud" shoots while preserving immature Arimbu shoots for future cycles. We analyzed the critical structure of the tea shoot—including the apical bud, fish leaf, and mother leaf—to understand why current machines often fail to distinguish these components, leading to a significant 35–40% yield loss and damage to maintenance foliage. These field insights provided the technical foundation for identifying the performance gaps in existing harvesters and establishing the criteria for our proposed precision harvesting improvements

2.1 Technical Specifications of Existing Tea Harvesting Machines

Table 01: Technical Specifications of Common Tea Harvesting Machines Used in Sri Lanka and Similar Plantation Systems

Machine	Power Source	Weight/ kg	Cutting Mechanism	Output / Efficiency	Figures
Kawasaki NV60H	Petrol (TU26 Mitsubishi engine, 2 Stroke, Air cool)	9.4 (Engine 6.2kg)	Shear blades (Blade length 600mm)	0.7–1.0 acre/day, ~15× faster than manual plucking	
Single Hand Electric Tea Harvester	Battery powered	1.8	Shear blades (Blade length 330mm)	30+ kg/hr (similar speed)	
Kawasaki Impeller machine	Battery powered	3.5	Shear blades with impeller/fan	Faster than manual, lower than NV60H	
Static Teeth Machine	Battery powered	1.75	Fixed teeth cutting system	Very low efficiency	
TRI selective harvesting shear	Manual	<1	Scissor mechanism	4.3 kg/hour, ~2× faster than hand plucking	

2.2 Performance Comparison of Tea Harvesting Machines

Table 02: Performance Evaluation of Tea Harvesting Machines Based on Tea Quality and Operational Limitations

Machine	Tea Quality Performance	Major Specifications	Main Limitations
Kawasaki NV60H	Low quality due to broken leaves, coarse leaves, and stem inclusion	High cutting speed, widely used	No selective plucking → harvests bud + coarse leaves together; high leaf damage; operator fatigue; post-sorting required
Single Hand Electric Tea Harvester	Moderate quality (non-selective)	Lightweight, portable	No bud-selectivity → mixed leaf harvest; battery dependency; limited coverage width; operator wrist fatigue during long use
Kawasaki Impeller machine	Moderate-low quality	Lightweight, improved handling	Leaf bruising due to impeller action; poor selectivity
Static Teeth Machine	Very poor quality	Simple mechanical design	Jamming in dense tea bushes; uneven cutting; high coarse leaf content; unsuitable for Sri Lanka terrain
TRI selective harvesting shear	High quality (closest to hand plucking)	Simple, low cost	Low productivity; fatigue; not suitable for steep slopes

2.3 Comparative Rating of Tea Harvesting Machines

Table 03: Comparative Performance Rating of Tea Harvesting Machines Based on Key Evaluation Criteria

Machine	Harvesting Efficiency	Tea Quality	Selectivity	Ergonomics (Operator Comfort)	Overall Rating (Average)
Kawasaki NV60H	5	2	1	1	2.25
Single Hand Electric Harvester	4	2	1	3	2.5
Kawasaki Impeller Machine	3	3	2	4	3
Static Teeth Machine	3	3	2	3	2.75
TRI selective harvesting shear	2	5	4	4	3.75
	"Excellent" -10x+ faster than manual; industrial-grade speed	"Excellent" Clean cuts; zero bruising; preserves leaf integrity	"Excellent" only mature shoots are harvested	"Excellent" <2kg; low vibration; ergonomic for 4+ hours	

Scale

- 5 = Excellent
- 4 = Good
- 3 = Moderate
- 2 = Poor
- 1 = Very Poor

3. Project Objectives & Methodology

3.1 Project Objectives

The main objective of this project is to develop an adaptive precision tea harvesting solution capable of selectively harvesting pluckable tea shoots while minimizing leaf damage and improving operational efficiency under Sri Lankan plantation conditions.

The specific objectives of the project are as follows:

Primary Objective

- To design a precision tea harvesting mechanism capable of selectively harvesting tea shoots based on the standard bud + 2 or 3 leaves plucking requirement.

Secondary Objectives

- To study and analyze existing mechanical tea harvesting systems currently used in Sri Lanka and identify their operational limitations.
- To evaluate the major challenges associated with non-selective harvesting, including leaf damage, coarse leaf inclusion, and poor adaptability to local terrain.
- To develop a lightweight (<2kg) and ergonomic harvesting mechanism suitable for steep and uneven tea plantation environments.
- To improve harvesting efficiency while maintaining tea quality comparable to manual plucking standards.
- To validate the proposed design through computer-aided modeling and engineering simulation.

3.2 Project Methodology

3.2.1. Methodology Workflow



Figure 04: Methodology Workflow

- **Literature Review / Problem Identification**

A literature review was conducted to study tea harvesting practices, plucking standards, tea shoot characteristics, and existing mechanical harvesting systems. A field visit to the Tea Research Institute (TRI) Sub Centre, Rathnapura was carried out to observe manual plucking, machine harvesting operations, tea shoot growth stages, and machine-induced damage to tea bushes.

Based on literature findings and field observations, major limitations of existing harvesting systems were identified, including non-selective plucking, damage to immature shoots, poor terrain adaptability, and operator fatigue. These findings were used to establish the main design requirements of the proposed system, including selective harvesting capability, lightweight structure, ergonomic operation, and reduced plant damage.

- **Requirement Analysis**

Based on the identified limitations of existing harvesting systems, the functional and design requirements of the proposed solution were defined. Key requirements include selective harvesting of pluckable shoots, preservation of immature shoots such as Arimbu, lightweight and ergonomic operation, adaptability to uneven terrain, and minimization of leaf and bush damage. These requirements were used as the primary criteria for concept generation and evaluation.

- **Concept Generation and Development**

Three preliminary design concepts were developed to address the identified limitations.

Concept A – Sensor-Guided Bending Shear Mechanism

- Uses stem bending behavior and sensor feedback to identify mature pluckable shoots before cutting.

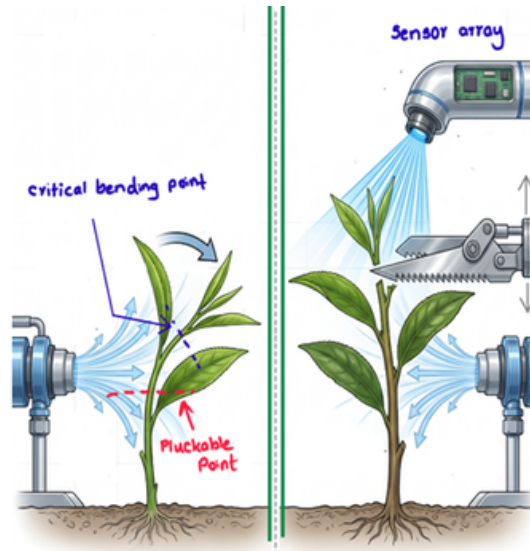


Figure 05: Concept A

Concept B – Optical Micro-Shear System

- Uses camera-based detection and leaf node counting to identify pluckable shoots and activate localized cutting.

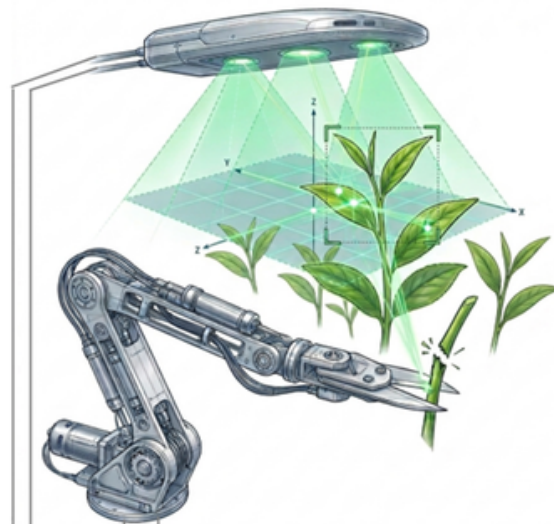


Figure 06: Concept B

Concept C – Stiffness-Based Selection Gate

- Uses mechanical stiffness differences between mature and immature shoots for passive sorting before cutting.

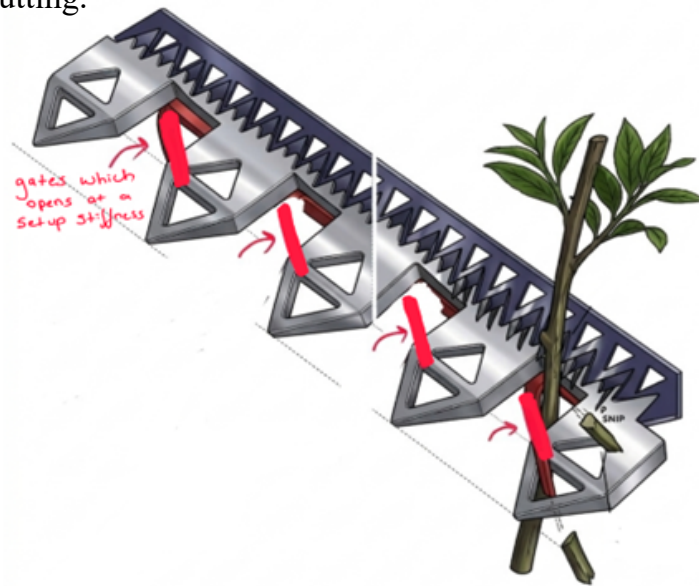


Figure 07: Concept C

• Concept Evaluation & Selection

The generated concepts will be evaluated using a decision matrix based on:

- selectivity performance
- simplicity of mechanism
- ergonomic suitability
- weight
- manufacturability
- terrain adaptability
- reliability
- cost feasibility

The most suitable concept, or combination of features from multiple concepts, will be selected for further development.

• Material Selection / Engineering Calculations

Preliminary engineering calculations were carried out to estimate key design parameters such as tea shoot dimensions, cutting force requirements, and basic loading conditions. Based on these requirements, suitable materials were selected by considering strength, weight, corrosion resistance, manufacturability, and cost effectiveness to ensure structural reliability and practical field applicability.

• 3D CAD Modeling / Simulation & Validation

The selected concept will be developed as a detailed 3D model and assembly to verify component arrangement, mechanism functionality, and overall dimensions. Engineering simulations will then be conducted to evaluate structural performance, stress distribution, deformation behavior, and design feasibility under expected operating conditions.

- **Comparative Evaluation**

The proposed design will be compared with existing tea harvesting systems based on key criteria such as selectivity, harvesting efficiency, tea quality preservation, ergonomic suitability, and terrain adaptability. This evaluation will be used to assess the effectiveness of the proposed improvements and justify the final design recommendation.

- **Final Design Recommendation**

The proposed design will be compared against existing harvesting systems based on:

- selectivity
- tea quality preservation
- ergonomics
- terrain adaptability
- operational feasibility

Final recommendations for adaptive precision harvesting improvements will be presented.

3.3. Software Tools

Table 04: Proposed Software Tools and Engineering Applications for Design, Analysis, and Documentation Phases

Software	Purpose
SolidWorks	3D modeling, assembly development, mechanism design
ANSYS	Structural simulation, stress and deformation analysis
MATLAB / Python	Data processing, calculations, analysis
Microsoft Excel	Data analysis, decision matrices, calculations

4. Project Planning & Timeline

Table 05: Project Timeline and Work Breakdown Schedule for the Adaptive Tea Harvesting System Development

	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Phase 1 – Research & Conceptualization												
Literature review	1											
Field visits/TRI data collection	1,2											
Requirements analysis		2										
Proposing concepts		2										
Phase 2 – Concept Selection & Detailed Design												
Concept selection			3									
Mechanism design			3,4,5									
Material selection				4,5								
CAD modeling (SolidWorks)				4,5,6								
Phase 3 – Simulation & Analysis												
FEA stress simulation (Ansys)							7,8					
Updating further developments							7,8,9					
Concept validation								8,9				
Phase 4 – Final Evaluation & Documentation												
Reporting										10,11		
Sustainability audit										10,11		
Final comparative assessment											11,12	
Presentation & submission											11,12	

5. Impact Analysis

5.1 Economic Considerations

The Sri Lankan tea industry currently faces a major labour shortage, increasing the need for efficient harvesting alternatives. The proposed precision harvesting system is expected to improve economic performance through targeted mechanization.

- **Reduced Harvesting Cost:** Mechanized harvesting can significantly reduce labour dependency and lower harvesting costs compared to manual plucking (manual harvesting costs approximately Rs. 375 per kg of made tea), improving overall estate profitability.
- **Improved Tea Market Value:** Selective harvesting of pluckable shoots is expected to reduce leaf damage, coarse leaf content, and refuse tea percentages, thereby improving final tea quality and market competitiveness.
- **Lower Post-Harvest Processing Cost:** Precise harvesting minimizes the collection of non-pluckable shoots, reducing sorting requirements and processing losses.
- **Local Adaptability:** Existing harvesting machines are mainly designed for temperate tea-growing regions and show poor adaptability to Sri Lankan terrain and plucking standards. The proposed system introduces a solution specifically adapted to local plantation conditions.

5.2 Social Considerations

The proposed system also contributes to workforce safety, labour productivity, and modernization of plantation operations.

- **Improved Occupational Safety:** The mechanized system reduces repetitive finger strain, hand injuries, and musculoskeletal fatigue associated with manual plucking.
- **Reduced Operator Fatigue:** Lightweight and ergonomic design considerations aim to reduce physical stress during prolonged harvesting, particularly on steep slopes.
- **Increased Worker Productivity:** Mechanized harvesting can improve daily harvesting capacity, potentially increasing worker earnings and operational efficiency.
- **Technology Adoption and Youth Engagement:** The integration of modern harvesting technology may encourage younger generations to participate in plantation-related work.

5.3 Environmental Considerations

Environmental sustainability is a critical factor in the design of the proposed harvesting system.

- **Preservation of Future Yield:** Selective harvesting protects immature Arimbu shoots, preserving future shoot development and maintaining yield continuity.
- **Reduced Plant Damage:** Controlled cutting minimizes deep cutting and unnecessary foliage damage, preserving maintenance leaves required for photosynthesis.

- **Cleaner Energy Utilization:** The adoption of battery-powered or energy-efficient mechanisms can reduce fuel consumption and environmental emissions compared to petrol-powered machines.
- **Long-Term Bush Health:** Reduced harvesting damage minimizes plant stress and supports sustainable tea bush productivity over repeated harvesting cycles.

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